

Bryan Ramirez-Gonzalez

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EDUCATION

The University of Southern California

Los Angeles, CA

Bachelor of Science in Computer Science, Merit Scholar, Honors Engineering Research Track

Expected May 2028

- Relevant Coursework: Data Structures, Algorithms, Object-Oriented Programming, Embedded Systems, Discrete Mathematics, Linear Algebra

SKILLS

Programming Languages: C, C++, Python, Java, JavaScript/TypeScript, SQL

Frameworks/Tools: LangChain, PyTorch, NumPy, Pandas, Flask, FastAPI, Matplotlib, React, Tailwind CSS, Firebase, GCP, Docker, AWS, Linux

EXPERIENCE

NVIDIA

Santa Clara, CA

Software Engineering Intern

May 2026 - September 2026

- Designed and implemented the pre-flight validation gate for an ISO 26262 FMEDA safety-report tool (slated for 5–10 yrs of production use), closing a class of silent failures via deterministic work-product checks + an LLM cross-check with human-in-the-loop review.
- Built a pytest-based regression suite (104 data-driven cases across 8 feature families) with golden-file Excel fixtures (openpyxl) and a pass/fail reporter, hardening a ~24-module verification pipeline and shipping it handoff-ready, reducing 80% of hardware engineers daily workload.

Hudson River Trading

New York, NY

Undergraduate Fellow (Inside HRT 2026 — Quant Research Track, 1 of 6 Selected)

May 2026

- Built an order-book matching engine and iterated from momentum, mean-reversion, and market-making strategies with a loss-limit kill switch.
- Outperformed the program's benchmark trader in the live reactive environment after analyzing Parquet market data using Pandas/NumPy.

Jane Street

New York, NY

Undergraduate Fellow (UNBOXED 2024; FOCUS 2025)

Jul 2024; May 2025

- Selected 1 of 37 (UNBOXED '24) and 1 of 14 (FOCUS '25); Wrote 150+ lines of SQL for data analysis and awarded a \$2,000 scholarship.
- Achieved 15% return improvement in market simulation games through 30+ hours of coursework in statistics and market-making.

USC Information Sciences Institute (ISI) — HUMANS Lab

Los Angeles, CA

Undergraduate Researcher — iOS-Agent: Open Research Platform for Physical Mobile GUI Agents

January 2026 - Present

- Built the first open research platform for autonomous AI agents on physical iOS, enabling a novel continuous-trajectory drag primitive and first autonomous multi-step drag on iPhone (tasks where Mobile-Agent, AppAgent, Mobile-Agent-v2 score near-zero) via 22+ tool primitives.
- Compressed 15-step agent tasks from 88s to 24s (3.5x speedup) via rolling-summary context compression (3.40x, 30% fewer tokens), direct HTTP to Maestro's XCTest runner (10–100x action-latency), and prompt caching (~90% per-step cost reduction).

Undergraduate Research Intern — LLM-assisted AI for TikTok Eating-Disorder Dataset (EDTok) [[arxiv](#)]

August 2024 - May 2025

- Published EDTok, an ethics-compliant, multimodal TikTok dataset of 43,040 videos and 577,071 comments (2019–2024), raising dataset precision by 24% using a two-stage filter: weak-supervision rules + LLM zero-shot relevance classification with prompt engineering.
- Surfaced platform-scale insights on dataset (537M views, 79.9M likes, 962k shares across 10.9k users) via longitudinal analysis, BERTopic topic modeling, and multi-label emotion classification, linking themes to affect (e.g., recovery to optimism/joy; body-image to fear/sadness).

ACTIVITY / EXTRACURRICULAR - 3x Hackathon Winner

- **Awards:** 2026 Claude Hackathon Winner, 2025 Caltech HackTech Winner, 2024 HackHarvard Winner, 2024 FTC Robotics Regional Semifinalist
- **Selective Programs:** D. E. Shaw Connect, Two Sigma New Seekers Summit, Susquehanna (SIG) Discovery Day, Datadog's DataPUPS

PUBLICATIONS - [[Google Scholar](#)]

- C. Bickham, **B. Ramirez-Gonzalez**, M.D. Chu, K. Lerman, E. Ferrara. [EDTok: A Dataset for Eating Disorder Content on TikTok](#), ICWSM 2025

PROJECTS

Basis (Award-Winning Agentic AI Workflow Automation Platform @ Fall 2025 USC LavaLab Winner — github.com/bryanrg22/Basis_Info)

Co-Founder & Lead Developer

September 2025 - May 2026

- Built a full-stack agentic workflow for cost segregation utilizing LangGraph, achieving 95% accuracy with 2 paying customers from top-5 firms.
- Implemented an IRS-grounded Agentic RAG exposed via an MCP tool registry, combining lexical BM25 (IRS-aware tokenization) and semantic search via FAISS working in parallel, while preserving structured tables intact (never chunked) and returning citation-backed results.

Lambda Rim (Quantitative Fantasy Sports Analytics Platform — github.com/bryanrg22/lambda-rim)

Lead Developer

June 2025 - May 2026

- Implemented a probabilistic forecasting pipeline: Poisson modeling (season λ), 100k-run Monte Carlo simulation, and GARCH(1,1) volatility forecasting for NBA player props, achieving 78% win rate and \$10 to \$3,000 profit growth deployed as a full-stack React/Flask application.
- Engineered automated data pipelines: NBA player stats (nba_api), daily betting lines (e.g. PrizePicks API via local cron), hourly injury reports (PDF web scraping via Cloud Scheduler + pdfplumber), and live game settlement (ESPN API), all feeding Firestore for real-time predictions.

Swerve (Award-Winning Agentic Procurement Platform Using LangChain Agents @ Caltech Hackathon Winner - github.com/bryanrg22/swerve)

Lead Developer

April 2025

- Built an event-driven LangChain agent that routes procurement queries through LLMs (intent classification → multi-step reasoning) to generate audit-traceable low-stock alerts and reorder recommendations; invited by Dryft to San Francisco Neo offices for post-hackathon collaboration.
- Implemented Flask APIs with CRUD to ingest CSVs and CADs into structured Firestore collections with Slack automation for real-time alerts